MEDIUM

GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

02

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

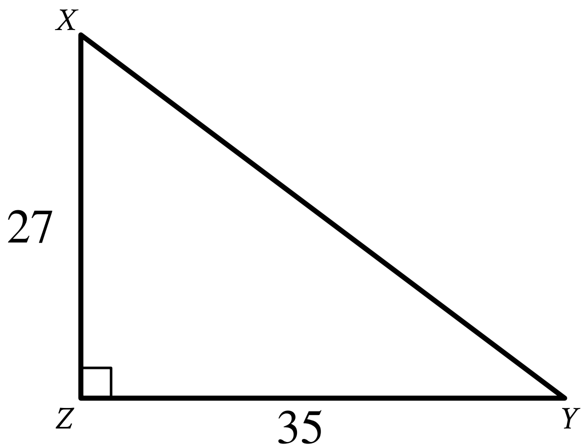
Q1 GRID-IN GEOMETRY AND TRIGONOMETRY

The number of radians in a 720-degree angle can be written as $a\pi$, where a is a constant. What is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Angle upper Z is a right angle.
- The length of side upper X upper Z is 27.
- The length of side upper Y upper Z is 35.
- A note indicates the figure is not drawn to scale.

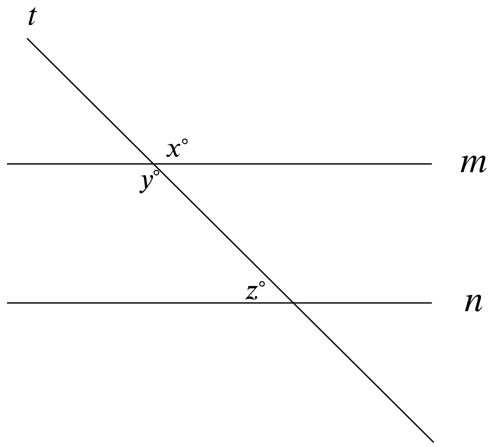
Triangle XYZ shown is a right triangle. Which of the following has the same value as $\sin X$?

A. $\tan X$

B. $\tan Y$

C. $\cos X$

D. $\cos Y$



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled t, m, and n.
- Line t intersects both line m and line n.
- At the intersection of line t and line m, 2 angles are labeled clockwise from top left as follows:
  - Top right: x°
  - Bottom left: y°
- At the intersection of line t and line n, 1 angle is labeled clockwise from top left as follows:
  - Top left: z°
- A note indicates the figure is not drawn to scale.

In the figure, lines m and n are parallel. If $x = 6k + 13$ and $y = 8k$ - 29, what is the value of z?

A. 3

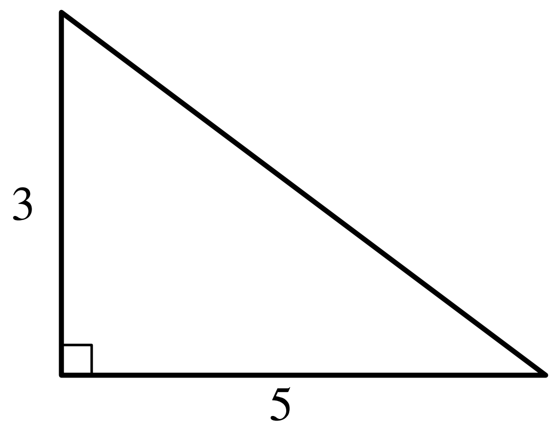
B. 21

C. 41

D. 139

Q4

GRID-IN GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- One angle is a right angle.
- The lengths of the 2 sides adjacent to the right angle are as follows:
  - 3
  - 5
- A note indicates the figure is not drawn to scale.

The figure shows the lengths, in inches, of two sides of a right triangle. What is the area of the triangle, in square inches?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

9 9 9 9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

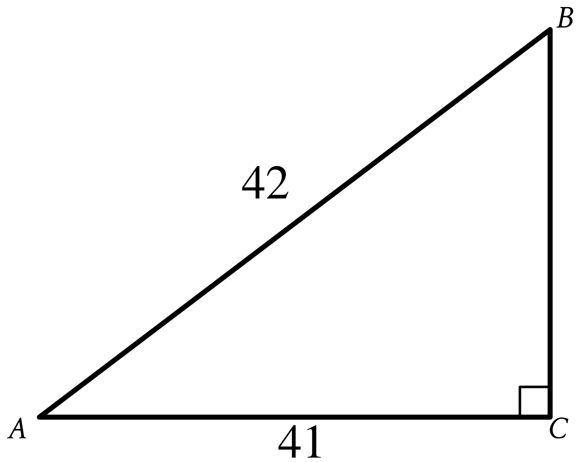
Q5 GRID-IN GEOMETRY AND TRIGONOMETRY

In the xy-plane, the graph of the equation $x - 3^2 + y - 5^2 = 9$ is a circle. The point $6, c$, where c is a constant, lies on this circle. What is the value of c?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Angle upper A upper C upper B is a right angle.
- The length of side upper A upper B is 42.
- The length of side upper A upper C is 41.
- A note indicates the figure is not drawn to scale.

What is the value of $\cos A$ in the triangle shown?

A. $\frac{42}{41}$

B. $\frac{41}{42}$

C. $\frac{1}{42}$

D. $\frac{1}{41}$

Q7 GEOMETRY AND TRIGONOMETRY

In triangle ABC, the measure of angle A is $54°$, the measure of angle B is $90°$, and the measure of angle C is $\frac{k}{2}°$. What is the value of k?

A. 36

B. 45

C. 72

D. 108

Q8 GEOMETRY AND TRIGONOMETRY

A right circular cylinder has a volume of 45π. If the height of the cylinder is 5, what is the radius of the cylinder?

A. 3

B. 4.5

C. 9

D. 40

Q9

GEOMETRY AND TRIGONOMETRY

The measure of angle R is $\frac{2\pi}{3}$ radians. The measure of angle T is $\frac{5\pi}{12}$ radians greater than the measure of angle R. What is the measure of angle T, in degrees?

A. 75

B. 120

C. 195

D. 390

Q10

GEOMETRY AND TRIGONOMETRY

In right triangle RST, the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin R$ is $\frac{\sqrt{15}}{4}$. What is the value of $\cos S$?

A. $\frac{\sqrt{15}}{15}$

B. $\frac{\sqrt{15}}{4}$

C. $\frac{4\sqrt{15}}{15}$

D. $\sqrt{15}$